ACS Photonics

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Article

Ultracompact Multicore Fiber De-Multiplexer Using an Endface-Integrating Graphene Photodetector Array

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Cite This: ACS Photonics 2022, 9, 1808−1813

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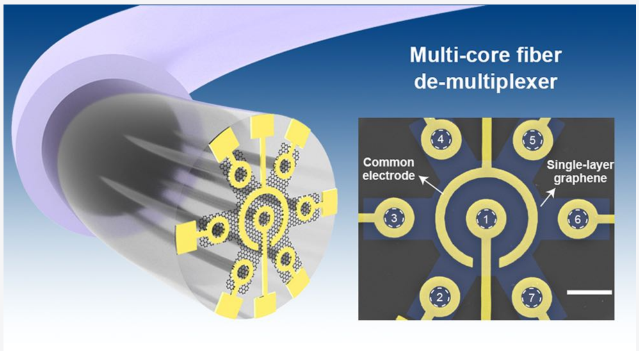
ABSTRACT: The use of multicore fiber (MCF) aims to increase the fiber density with downsized fiber systems, which is constrained by a de-multiplexer with a large footprint that converts the MCF into numerous single-mode fibers and corresponding number of infrared photodetectors. In this study, we demonstrate an ultracompact seven-core fiber de-multiplexer by integrating a patterned single-layer graphene (SLG) photodetector array on a single fiber endface. An optimized electrode configuration was successfully fabricated with a strongly asymmetrical structure on patterned SLG. Owing to the efficient photocarrier separation at the built-in electric field across the metal-doped junction in graphene, we realized self-powered seven-core photodetection using the device. Remarkably, our device exhibits a coupling-free, miniaturized device volume with zero power consumption, a high transmittance, and low manufacturing cost. This device downsizes traditional bulky receiver modules to the micrometer scale with an atomical thickness, which may inspire the development of next-generation highly integrated optical and optoelectronic systems.



KEYWORDS: *multicore fiber, optical fiber device, graphene, photodetector*

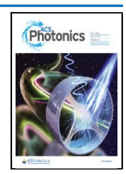
■ INTRODUCTION

Multicore fiber (MCF) is an essential space-division multiplexing solution used to increase the fiber density with downsized fiber systems, which have been widely studied for high-capacity optical communication,[1−3] sensing,[4−10] imaging,[11,12] and other applications.[13] However, a conventional de-multiplexer module in an MCF system requires the conversion of MCF into several single-mode fibers (called fan-in/fan-out modules), which are then connected to the corresponding number of infrared photodetectors, resulting in a complex system, large footprint, high power consumption, and high cost. To solve this problem, integrated MCF devices have been investigated, including on-chip grating coupler arrays to replace fan-in/fan-out modules[14] and two-dimensional (2D) photodetector arrays on chips as compact terminal photoreceivers.[15] However, these approaches face the challenge of improving the coupling efficiency and alignment tolerance between optical fibers and chips, and a low-loss, in-line analysis in MCF systems has yet to be conducted.

As a burgeoning approach, the integration of optoelectronic materials, particularly atomically thin 2D materials, onto optical fibers has recently become an effective way to overcome the material limitations inherent to silica optical fibers. Similar to on-chip configurations, fiber endface optoelectronic devices have been successfully demonstrated with diverse materials and device structures.[16,17] For instance, 2D materials and their stacked heterostructures, which serve as photosensing materials, have been integrated on fiber endfaces, D-shaped fibers, and microfibers for high-performance photodetection.[18−22] However, state-of-art fiber-compatible fabrication techniques impede the realization of complex electrode structures on optical fibers, thereby limiting their further applications. For special optical fibers, including MCFs, the preparation process standardized for planar substrates needs to be optimized to manufacture complex electrode structures on the tiny endface of a single fiber.

Herein, we designed and fabricated an ultracompact seven-core fiber (SCF) de-multiplexer by integrating a chemical vapor deposition (CVD)-grown single-layer graphene (SLG) photodetector array on a fiber endface using a customized electron-beam lithography (EBL)-based technique that overcomes the unconventional shape of the optical fibers with good manufactory stability and repeatability (see the Supporting

Received: March 6, 2022
Published: May 3, 2022



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ACS Photonics 2022, 9, 1808−1813

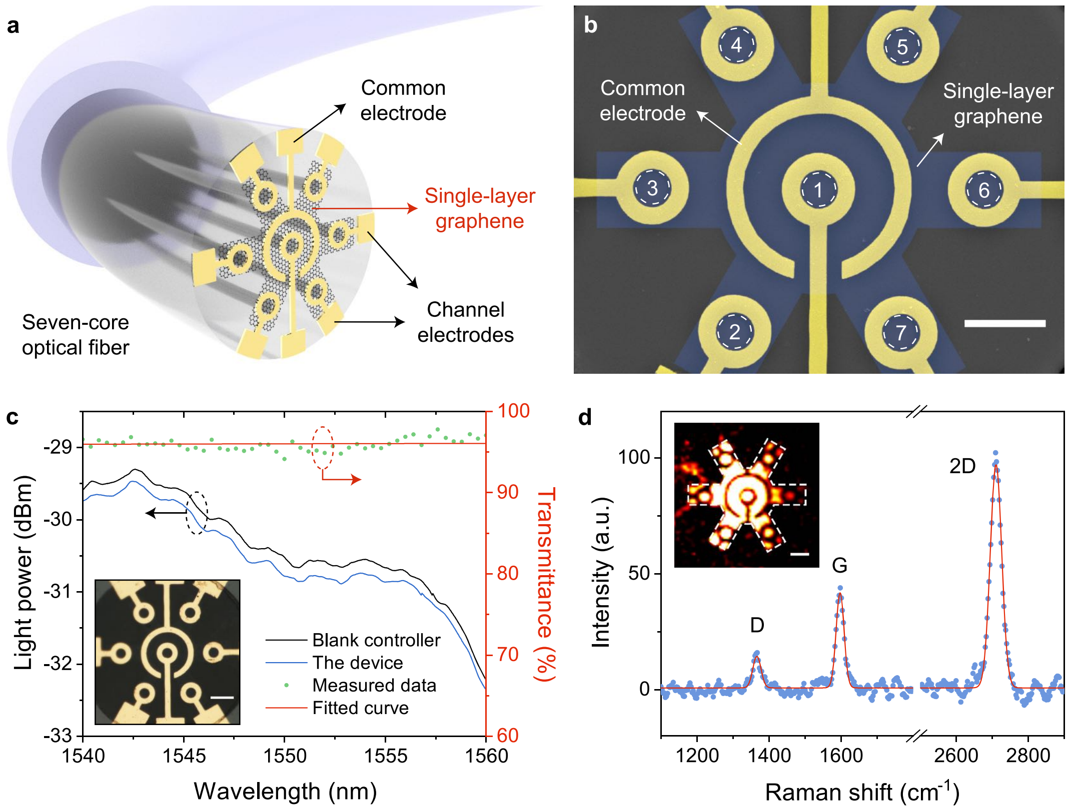

Figure 1. Device structure. (a) Schematic illustration of the integrated graphene photodetector on a commercial seven-core optical fiber. (b) Pseudo-color SEM image of the device. The yellow area presents the Au/Ti electrodes, including a common electrode and seven channel electrodes. The blue area presents the patterned SLG film. The areas in white dashed circles present fiber cores, which are covered by an SLG film and surrounded by channel electrodes. The scale bar is 20 μm. (c) Optical loss of the device. The black curve is the reference spectrum, whereas the blue curve is the light power transmittance from the device. The green dots are the measured transmittance data, which are linear fitted, as shown by the red line. Inset: microscopic image of a prepared device, where the patterned SLG is nearly invisible. The scale bar is 20 μm. (d) Raman spectra of the SLG using a 488 nm laser. Inset: 2D Raman intensity mapping showing the patterned SLG in the white dashed area. The scale bar is 20 μm.

Information for details). The electrode configuration was significantly simplified using seven channel electrodes and one common electrode, which showed a strongly asymmetrical structure. The efficient photocarrier separation at the strong built-in electric field across the metal-doped junction in the SLG enables the self-powered photodetection of seven individual cores by the device. Owing to the high transparency of the SLG, the device showed a transmittance of over 96%, benefiting the in-line, real-time light status analysis. The miniaturized structure of the device that downsizes traditional bulky fan-out module and numerous photodetectors to the micrometer scale with an atomical thickness may pave the way for highly integrated next-generation optical and optoelectronic systems with diverse applications.

■ RESULTS

Device Design. The designed device, schematically shown in Figure 1a, has eight electrodes, that is, one common electrode and seven channel electrodes, prepared on a patterned SLG film above a flat SCF endface. The scanning electron microscopy (SEM) image in Figure 1b shows the electrode configuration of the prepared device. Every fiber core (white dashed area) was surrounded by a ring-like channel electrode, forming pairs of asymmetrical electrodes together with the common electrode. The ring-like channel electrodes will not block the light from the fiber cores, thus obtaining high transparency of the device. The patterned SLG, which serves as a photosensing material, covers all fiber cores, and connects the seven channels. A microscopic image of the prepared device is shown in Figure 1c, where the patterned SLG is nearly invisible owing to weak light absorption, showing a transmittance of over 96%. To identify the existence of the patterned graphene, we obtained the in situ Raman spectra, as shown in Figure 1d, where three narrow and symmetric characteristic peaks were observed at 1367 cm^{-1} (D-band), 1598 cm^{-1} (G-band), and 2709 cm^{-1} (2D-band). The peak intensity ratio of the 2D-band to the G band was approximately 2, implying that the material was SLG.[23,24] With the assistance of the Raman intensity mapping shown in the inset of Figure 1d, the profile of the patterned SLG area (white dashed area) can be clearly seen. The details of the material preparation and device fabrication are demonstrated in the Methods and Supporting Information section.

Self-Powered Photosensing Performance. The *I*−*V* characteristic curves of the central channel and a typical outer channel, collected using a digital source measure unit (Keithley SMU 2450, Tektronix), are shown in Figure 2a. The linearity of the *I*−*V* curves suggests an Ohmic contact between the Au/Ti electrodes and graphene. The inset of Figure 2a shows the *I*−*V* curves with and without 1550 nm light illumination. The generated photocurrent (defined as $I_p = I_{light} - I_{dark}$) was nearly independent of the bias voltage. This is because the external

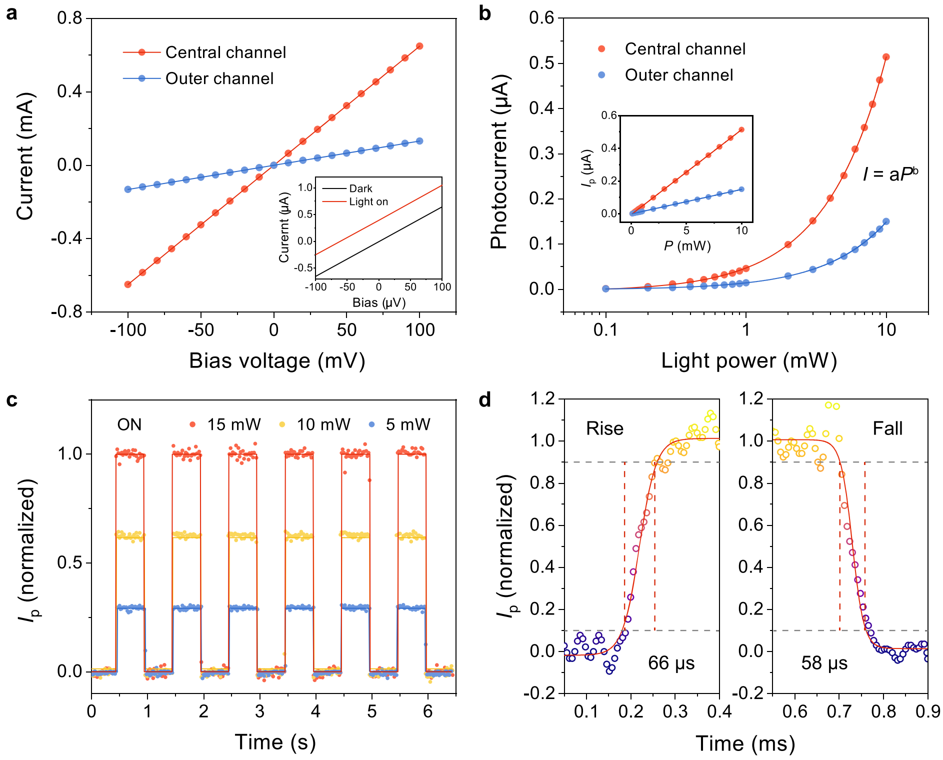

Figure 2. Self-powered photodetection performance (λ = 1550 nm). (a) $I-V$ characteristic curves of the core channel (red) and a typical outer channel (blue). Inset: black and red lines are the $I-V$ curves of the core channel in a dark environment and under illumination. (b) Spontaneous photocurrents generated in the core channel (red) and outer channel (blue) as a function of illuminating power in a semi-log coordinate. Inset: the relationship of spontaneous photocurrent and power in a linear coordinate. (c) View of the normalized self-driven photocurrent dynamics during several cycles of light modulation of the device under different light powers. (d) One cycle of photocurrent dynamics under zero bias, showing the measured rise/fall time of the device.

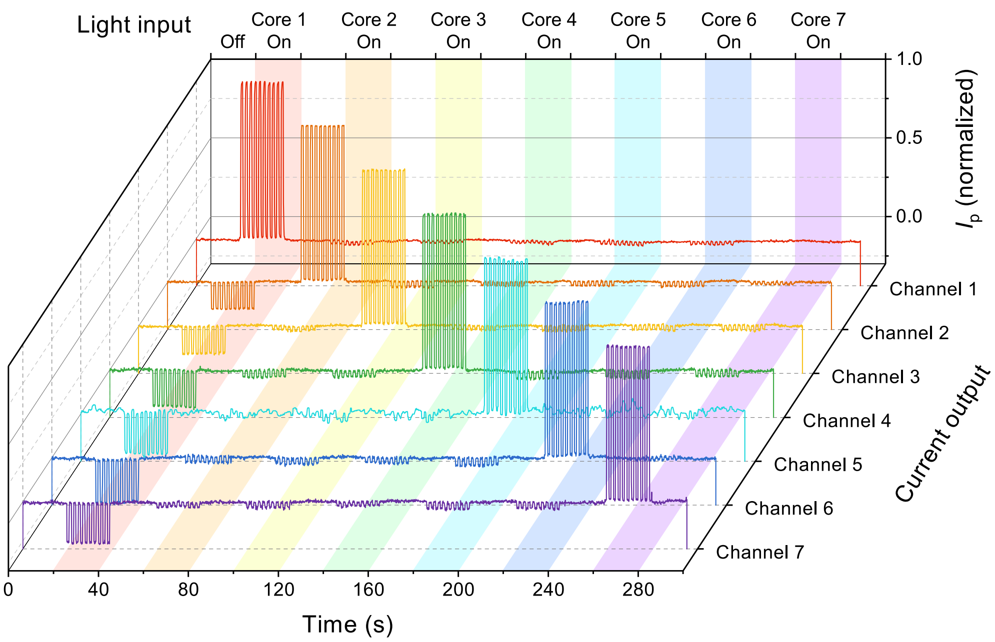

Figure 3. Seven-core photodetection. The figures from top to bottom are the collected photocurrent outputs (normalized) from channels 1−7, measuring the light response when cores 1−7 are illuminated individually. The input light is modulated with a 0.5 Hz on-off frequency, switching the status of the blocking (white area) and opening (colored area) every 20 s.

bias has a limited effect on the generation and separation of the photocarriers but will result in a high dark current. Therefore, we focused on the photodetection performance of our specially designed device under a zero bias. As shown in Figure 2b, the spontaneous photocurrents generated in the central and outer channels show exponential growth relations of light power (P) in semi-log coordinates. Because the graphene area in the central channel is larger than that in the outer channel, it has a

https://doi.org/10.1021/acsphotonics.2c00367
ACS Photonics 2022, 9, 1808−1813

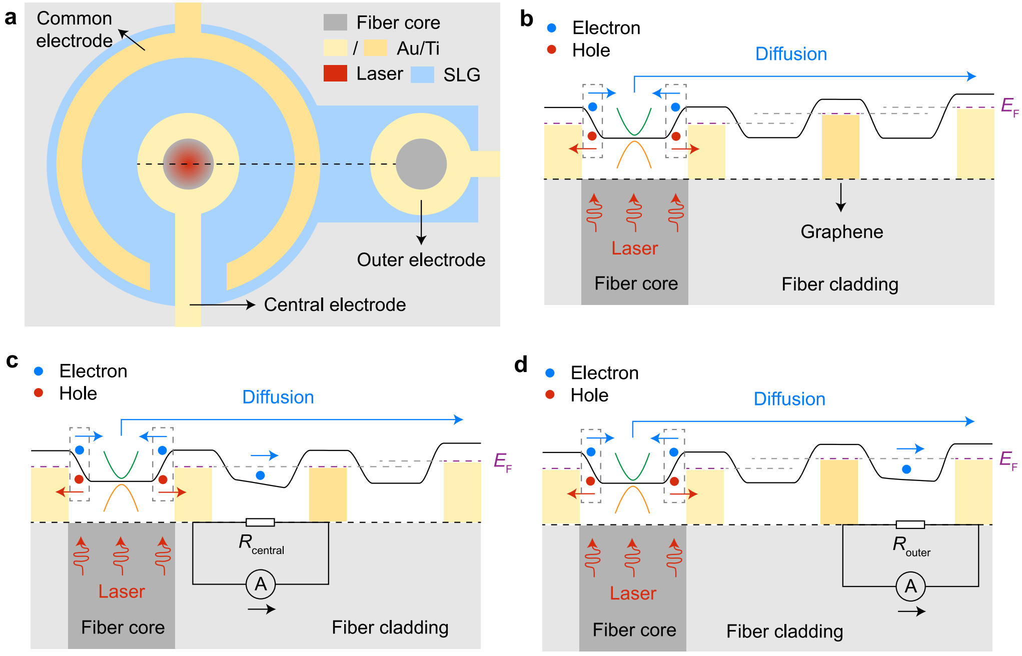

Figure 4. Schematic of the spontaneous photocurrent generation mechanism. (a) Schematic illustration of a simplified device including a central channel and an outer channel. The central core is illuminated. Cross-sectional potential profile of the device along the black dashed line in (a) is shown under open-circuit condition (b), short-circuit condition in the central channel (c), and outer channel (d). In both (b−d), the solid black line represents the potential within the graphene, the dashed purple line shows the Fermi level (EF), the dashed gray line is a fiducial line illustrating the EF variation across the channel, the dashed rectangular boxes denote the metal−graphene interface, and the green and orange parabolic curves represent a graphene band structure.

smaller resistance and exhibits a larger photocurrent under the same illumination power. The photoresponsivity (R), which is defined as $R = I_p/P$, is one of the most important parameters of a photodetector. Here, we measured an R of 35.4 μA/W at a light power of 10 μW, without bias. In general, owing to the large dark current and high noise of pristine graphene devices, it is difficult to distinguish the signal from the noise when the illumination power is less than approximately 1 mW. However, for our asymmetrically designed device, which can operate without an external bias voltage, the noise and dark current are strongly suppressed, thus reducing the detection limit. Moreover, the R value can be further improved by bringing in other efficient photogeneration effects (e.g., forming a heterostructure[18]) or enhancing the light absorption of graphene (e.g., building resonant structures[25]).

The response time (τ) is also a crucial figure of the merit for evaluating the performance of a photodetector. The self-driven photocurrent dynamics under periodically switched light at three different illumination intensities are shown in Figure 2c. The currents increase under illumination and recover when the light is turned off, with a fast response to the illumination switch. The 10−90% rise/fall time of the device was measured as ∼60 μs, as shown in Figure 2d. Notably, the instrumentation limits faster measurements of the device, and the actual response time may be faster because of the ultrafast carrier dynamics for carriers inside graphene, which shows an intrinsic bandwidth potential of over 500 GHz.[26] The long-time optical switching test of the device is performed in Figure S6 in the Supporting Information, and the device shows good stability and repeatability.

Seven-Core Photodetections. Because a common electrode is used to collect photocurrents from the seven channels, and the patterned graphene in the device is interconnected between the channels, it is important to investigate the channel crosstalk. As shown in Figure 3, we measure the light response of seven channels to incident light from seven individual cores, which are illuminated one after another. The illumination with a 0.5 Hz on-off frequency is modulated using an electro-optical modulator (LN81S-FC, Thorlabs), which is driven by a signal generator (33522A, Agilent). For the central channel, a significant photocurrent is observed when illuminating its core and the collected currents show a much lower negative value when the outer cores are illuminated, realizing an unambiguous photodetection. For an outer channel, a significant photocurrent is observed when illuminating its own core, and the crosstalk currents are much lower when illuminating the other outer cores. However, the collected current from the outer channel when illuminating the central core is not negligible, showing a negative value proportional to the light power, which can be further eliminated through a precalibration (see Figure S5 in the Supporting Information). Although the channel 2−7 are relatively symmetrical in positions, the photoresponses are different, which is mainly due to the slight deviation of electrodes during fabrication and the inhomogeneity of graphene.

The mechanism for the generation and collection of the photocarriers is shown schematically in Figure 4, taking an example of illuminating the central core. When light is incident in the fiber core area, photoexcited electron−hole pairs are generated and separated at the graphene−metal interface, where there exists a strong built-in electric field across the metal-doped graphene and the bulk graphene region.[27] In the open-circuit case (Figure 4b), photogenerated holes are injected into the electrodes near the illumination area (central channel electrode), whereas electrons diffuse into the bulk

https://doi.org/10.1021/acsphotonics.2c00367
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graphene away from the illumination area, resulting in a difference in the doping level of the graphene. The photogenerated carriers do not produce a current in the external circuits but produce an open-circuit voltage (V_{OC}), whose value is determined by the doping difference in the graphene. Consequently, the V_{OC} value of the central channel is larger than that of the outer channel owing to the diffusion process. In the short-circuit case, an ammeter was used to connect the electrode pairs (Figure 4c,d), and a photocurrent $I_{ph} = V_{OC}/R_g$ was generated in the external circuit, where R_g is the resistance of the graphene channel. The resistance of graphene in the central channel ($R_{central}$) is smaller than that in the outer channel (R_{outer}), with a larger V_{OC} between the central channel, where the generated photocurrent is much larger than crosstalk current to the outer channel. This also explains that when measuring the photocurrent of an outer channel, the crosstalk photocurrent from the center channel (illuminating the center core) is more obvious than that from other outer channels (illuminating other outer cores).

Because of the large difference between the photocurrent generated by the channel itself and the crosstalk currents generated by other channels, when multiple channels are illuminated with a similar light power, it is possible to directly recognize the on/off state of each channel by measuring the corresponding spontaneous photocurrents, realizing simple seven-channel photodetection. For the unambiguous photodetection of the illumination intensity of each fiber core, a seven-variable linear equation group must be solved by establishing the photocurrent-power relationships of each channel under the illumination from seven individual cores.

CONCLUSIONS

In summary, we demonstrated an ultracompact SLG array-based endface-integrated MCF demultiplexer. The improved design of the electrode array promises a fast dynamic response at a zero-bias operation, thereby allowing low power consumption. Moreover, a variety of 2D materials with photoresponses to infrared light can be potential candidates to replace graphene in this work, such as black phosphorus, Bi_2Se_3, $MoTe_2$, and so on. This all-fiber photodetector, with the combined advantages of a miniaturized device volume, high transparency, zero power consumption, and low cost, opens the door to high-performance, highly integrated optical and optoelectronic systems.

METHODS

Device Fabrication. The sequential fabrication process of the device is shown in Supporting Information, Figure S1. (i) A commercial seven-core optical fiber was cleaved to obtain a flat endface. (ii) CVD-grown SLG on the copper substrate was transferred onto the fiber endface by coating a layer of polymethyl methacrylate (PMMA), dissolving the Cu substrate, and removing PMMA through acetone after the transfer.[18,19] (iii) An electron resist layer of PMMA, and a layer of water-soluble polyaniline (AR-PC 5090) were spin-coated step-by-step onto the fiber endface using a customized module. (iv) EBL was used to pattern the protective PMMA layer on the SLG. (v) The SLG layer was patterned using O_2 plasma etching to remove the area without protection from PMMA. (vi) The PMMA layer was removed using acetone after transfer. (vii) Three electron resist layers, consisting of methyl methacrylate, PMMA, and water-soluble polyaniline, were spin-coated on the fiber endface sequentially (see Supporting Information, Figure S2). (viii) The EBL process was used to pattern the electrodes. (ix) Next, 5 nm Ti and 50 nm Au were evaporated onto the fiber endface one after the other. (x) Finally, electrodes were prepared on the SLG after a lift-off procedure was conducted. Our standard fabrication method shows the potential of developing a complementary metal oxide semiconductor compatible procedure on optical fibers.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsphotonics.2c00367.

Device fabrication procedures, details of the EBL process on the SCF endface, device calibration demonstration, and device stability measurement (PDF)

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https://doi.org/10.1021/acsphotonics.2c00367
ACS Photonics 2022, 9, 1808−1813

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https://pubs.acs.org/10.1021/acsphotonics.2c00367

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Funding

National Key R&D Program of China (2021YFA1401103 and 2018YFA0305800), the National Natural Science Foundation of China (61925502, 51772145, and 6210031520), Jiangsu postdoctoral research funding program (2021K228B), and Natural Science Fouding of Jiangsu Province (BK20211014).

Notes

The authors declare no competing financial interest.

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